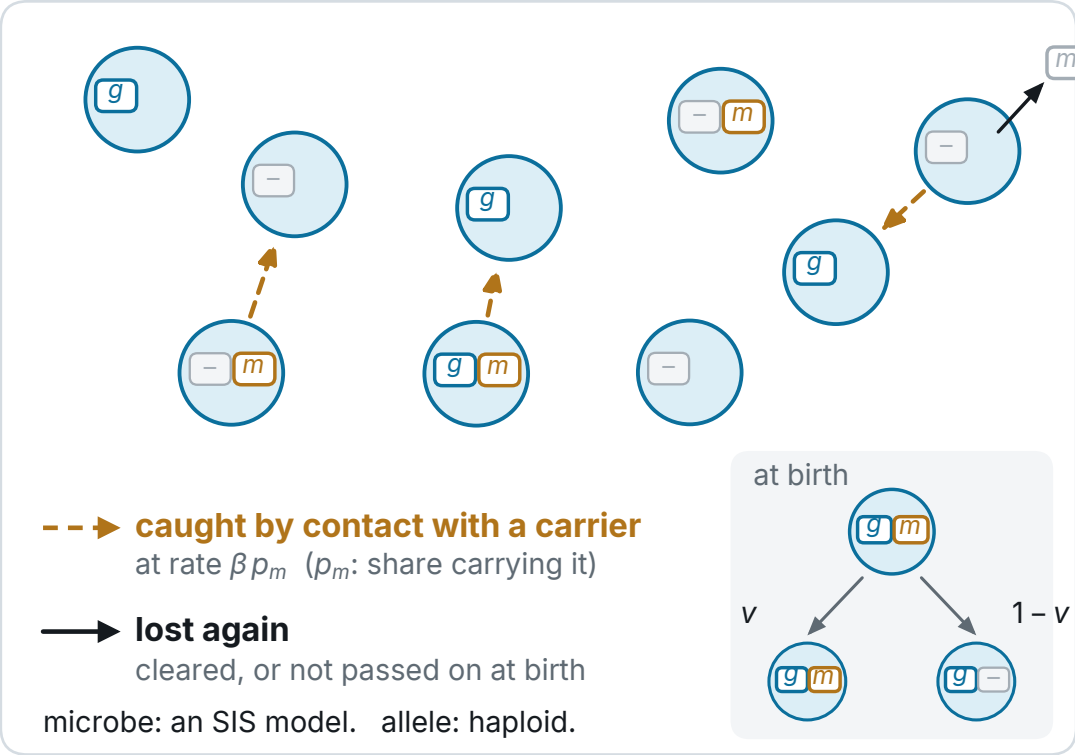
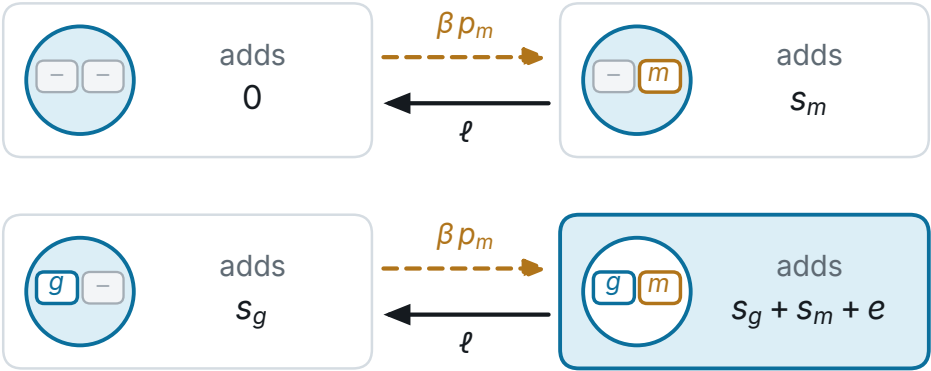


A host allele that is inherited, a microbe that is caught and lost



Four kinds of host



what each kind adds to a growth rate that is otherwise negative

host fitness =
$$\underbrace{\boxed{g}}_{s_g \text{ the allele}} + \underbrace{\boxed{m}}_{s_m \text{ direct effect}} + \underbrace{\boxed{g} \times \boxed{m}}_{e \text{ interaction}}$$

the microbe can help directly, or only together with the allele